MILITARY SPECIFICATION

NUT, SELF-LOCKING, SPANNER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for self-locking aluminum nuts used to secure fuel system components.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

GGG-W-665	Wrench, Spanner
PPP-H-1581	Hardware (Fasteners and Related Items) Packing of
PPP-T-76	Tape, Packaging, Paper (for Carton Sealing)
VV-P-236	Petrolatum, Technical

MILITARY

MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-I-6866	Inspection, Penetrant Method of
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5310

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STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standards for Federal Services
FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-1312	Fasteners, Test Methods

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

American National Standards Institute

ANSI B46.1	Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. The nut shall be fabricated from material as specified on the applicable specification sheet.

3.3 Construction. Nuts shall be of the prevailing torque-type. The locking device shall not operate by means of separate movement as a result of installation and shall not depend upon pressure on the bearing surface for the locking action.

3.3.1 Bearing surface. The bearing surface shall be normal to the axis of the pitch diameter of the thread within the values specified on the applicable specification sheet. The surface roughness of the bearing surface shall be not greater than 125 microinches Ra in accordance with ANSI B46.1.

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3.3.2 Threads. Unless otherwise specified, thread dimensions shall be in accordance with MIL-S-8879. Thread gaging shall be performed after anodic coating in accordance with the applicable thread specification (except that the nut with locking device incorporated but without lubricant shall allow the go plug gage members to enter not less than three-fourths turn before engagement of the locking device). When lubricant prevents use of standard gages, the nut shall permit free rotational (finger torqued) bolt engagement of at least three-fourths turn.

3.3.3 Surface treatment. The nuts shall be anodized in accordance with MIL-A-8625 Type II, Class 2, color red, color number 11136 in accordance with FED-STD-595.

3.3.4 Locking element. The locking element shall be non-metallic and resistant to JP-4, JP-5 and JP-8 when tested in accordance with 4.4.7. There shall be not more than two elements and if there are two, the elements shall be spaced not more than 120 degrees apart and shall meet the torque requirements of paragraph 3.5.2.

3.4 Tensile strength. The nuts shall have a tensile strength not less than the values specified in Table I when tested as specified in 4.4.2.

TABLE I. Nut characteristics.

Nominal dia.	1 Ultimate tensile strength lb. min.	2 Wrench torque in. lb. min.	Discontinuities in. max.
1.75	29000	1575	.066
2.00	33000	1800	.075
2.50	42000	2250	.094
2.75	46000	2475	.104
3.00	50000	2700	.113
3.25	54000	2925	.121
3.50	59000	3150	.131

NOTES:
1. The required ultimate tensile strength (Ftu) is 75% of the value obtained when calculated by Ftu = stress area (As) x ultimate shear strength (Fsu). As = Pi (E)(L_E)/2 from FED-STD-H28, appendix A5, sec. 3, where E is nominal pitch diameter of external thread and L_E is the length of engagement. F_{su} = 36000 psi (for aluminum 2024 T851).

2. Wrench torque is friction factor (K) x nominal diameter (D) x load (L in lbs) or KDL. K = .3, L = 3000.

3.5 Torque.

3.5.1 Wrench torque. The nuts shall withstand the wrench torque when tested as specified in 4.4.3. Upon completion of the test, the nuts shall show no indications of permanent deformation which may prevent its proper

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installation and removal with a wrench conforming to GGG-W-665. The nuts shall show no evidence of cracking when tested in accordance with 4.4.3.

3.5.2 Torque effectivity. The locking torque shall be not greater than 200 inch pounds and the breakaway torque shall be not less than 70 inch pounds when tested as specified in 4.4.4.

3.5.3 Permanent set. The locking torque shall be not greater than 200 inch pounds and the breakaway torque shall be not less than 70 inch pounds when tested as specified in 4.4.5.

3.6 Discontinuities. Discontinuities in the nuts shall be not greater than the depth limitations shown in Table I when inspected as specified in 4.4.6.

3.6.1 Cracks. Care shall be exercised not to confuse cracks with discontinuities. Cracks are not permitted in any location.

3.6.2 Laps, seams and inclusions. Any laps, seams and inclusions in the nuts shall be in accordance with the requirements of Table I.

3.7 Identification marking. Each nut shall be numbered and marked in accordance with the applicable specification sheet and include the manufacturer's trademark or identification.

3.8 Workmanship. Workmanship shall be consistent with the type of product, finish and the class of thread fit specified in the applicable specification sheet. Sharp edges shall be broken; hanging burrs and slivers shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as Quality Conformance Inspection (see 4.3).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable test method document or applicable paragraph(s) in the specification.

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4.3 Quality conformance inspection.

4.3.1 Sampling for examination. Test samples shall be selected at random in accordance with MIL-STD-105, inspection level and acceptance quality level (AQL) as specified in Table II. Identical sample items may be used for any of the tests, provided selection of random samples is maintained and known characteristics of the sample are not used to influence the integrity of test results.

TABLE II. Quality conformance inspection.

Test	Reference paragraph	MIL-STD-105 sampling level	AQL %
Examination of product	4.4.1	S3	2.5
Axial strength	4.4.2	S1	1.0
Wrench torque	4.4.3	S1	1.0
Torque effectivity	4.4.4	S1	1.0
Permanent set	4.4.5	S1	1.0
Discontinuities	4.4.6	S3	1.5
Chemical resistance	4.4.7	S1	1.0

4.3.2 Lot. A lot shall consist of finished nuts of the same type and diameter fabricated by the same process, from one heat of material, heat treated in the same manner and produced as one continuous run.

4.3.3 Test conditions. Unless otherwise specified, either herein or in the contract, tests shall be conducted at ambient conditions.

4.4 Examinations and tests.

4.4.1 Examination of product. The nuts shall be examined for conformance to this specification and applicable specification sheets with respect to material, workmanship, dimensions, design and construction and finish.

4.4.2 Axial strength. Nuts shall be tested in accordance with MIL-STD-1312, Test 8. The nuts shall support the minimum ultimate axial load listed in Table I without failure.

4.4.3 Wrench torque. The nuts shall be installed on a test mandrel as specified on Figure 1 and torqued to the wrench torque test values specified in Table I. Wrenches shall conform to GGG-W-665. Deformation which interferes with the proper application and removal of the nut with the wrench shall be cause for rejection.

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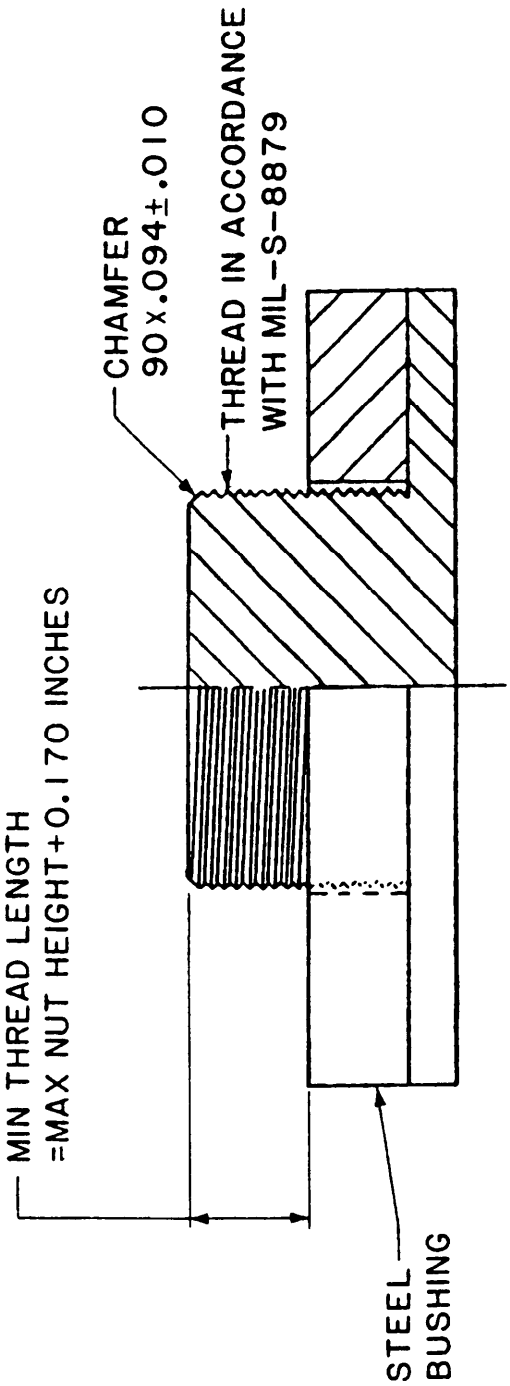


FIGURE 1. Test mandrel.

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4.4.4 Torque effectivity. The nuts shall be installed on an alloy steel test mandrel as specified on Figure 1. Each nut shall be screwed on and off the mandrel a total of 15 consecutive installations and removals. Petrolatum jelly in accordance with VV-P-236 shall be used as a lubricant and applied to the test mandrel on the first, fifth and tenth cycles of installation and removal. The nut shall be considered fully installed when a length of two thread pitches including chamfer, extends beyond the nut and fully removed when the locking device is disengaged. Readings of the maximum locking torque and breakaway torque shall be obtained on the first, seventh and fifteenth installations and removals. Breakaway torque shall be the minimum torque required to start removal of the nut from the installed position and shall not be less than 70 inch pounds. The maximum torque shall be the maximum torque value during the 15 installation and removals and shall be not greater than 200 inch pounds.

4.4.5 Permanent set. Permanent set shall be evaluated by subjecting the nut to one complete installation and removal cycle on a maximum mandrel whose pitch diameter +.0.000 -0.0004 below the maximum pitch diameter and then repeating the cycle with the same nut on a minimum mandrel specified on Figure 1.

4.4.6 Discontinuities. The indication of discontinuities shall be determined by penetrant inspection in accordance with MIL-I-6866, Type I. Inspection shall be performed on finished nuts, but prior to lubrication. Indications in themselves shall not be cause for rejection. Sample nuts indicating defects shall be sectioned and measured microscopically to determine conformance to the requirements specified in 3.6.

4.4.7 Chemical resistance. The chemical resistance of the material used for the locking element shall be tested by soaking finished nuts in JP-4, JP-5 and JP-8 for a continuous period of 72 hours at a temperature of 130 ± 5°F. Immediately after the soak period, a sharp instrument shall be used to attempt to pry the element away from the metal surface of the nut. Failure of the element to adhere to the nut shall be cause for rejection. Following this examination, the nut snall be tested in accordance with 4.4.3 and 4.4.4. Failure to meet the requirements of these tests shall be cause for rejection.

4.5 Preservation, packing and marking. Shipment of nuts shall be examined for conformance as specified in Section 5.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation, packing and marking shall be as specified in PPP-H-1581.

5.2 Marking of shipments. In addition to any special marking required by the contract or purchase order, unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The nuts covered by this specification are intended for use in aircraft fuel tanks to secure fuel lines.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Specification Sheet Part Number.

c. Quantity.

d. Selection of applicable preservation and packing in accordance with PPP-H-1581.

6.3 Definitions.

6.3.1 Crack. A crack is a clean crystalline break passing through the grain or grain boundary without the inclusion of foreign elements.

Custodians:
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Preparing activity:
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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-N-85754	Nut, Self-Locking, Spanner

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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